

Fred Hohman

Research Scientist (*HCI* + *ML*)

I design and develop **interactive interfaces** to help people **understand machine learning** models and data-driven systems. Besides building tools, I also create data visualizations and write interactive articles to simply communicate complex ideas.

I have collaborated with researchers, designers, developers, and artists while working at Apple, Microsoft Research, NASA Jet Propulsion Lab, and Pacific Northwest National Lab.

✉ fredhohman@apple.com
🏠 fredhohman.com
📄 CV PDF

🐦 @fredhohman
🌐 @fredhohman
🎓 Google Scholar

Education

Aug. 2015 — Dec. 2020

Ph.D. in Computational Science & Engineering

Georgia Institute of Technology, Atlanta, GA

Thesis: *Interactive Scalable Interfaces for Machine Learning Interpretability*

Advisor: Duen Horng (Polo) Chau, Co-advisor: Alex Endert

Committee: Duen Horng (Polo) Chau, Alex Endert, Chao Zhang, Nathan Hodas, Scott Davidoff, Steven Drucker

🏆 ACM SIGCHI Outstanding Dissertation Award

🏆 Georgia Tech College of Computing Dissertation Award

🔗 Thesis Project Page 📄 Thesis 🎥 Thesis Recording

May 2018

M.S. in Computational Science & Engineering

Georgia Institute of Technology, Atlanta, GA

GPA: 4.00/4.00

Aug. 2011 — May 2015

B.S. in Mathematics, B.S. in Physics

University of Georgia, Athens, GA

Thesis: *3D Printing the Trefoil Knot and its Pages*

Overall GPA: 3.84/4.00, Magna Cum Laude, Area of Emphasis in Applied Mathematics

🔗 Thesis Project Page 📄 Thesis

Industry Research Experience

Jan. 2021 — Present

Apple, Seattle, WA

Research Scientist, Design + Visualization Group

Research and design for everyone in the machine learning process—the developers who build models, the designers who create experiences, and the people who use technology.

Summer 2019

Apple, Seattle, WA

Research Intern, Turi Human-centered Machine Learning Group

Mentor: Kanit Wongsuphasawat, Kayur Patel

Designed and developed interactive visualizations for data iteration in machine learning, published at CHI 2020.

Summer 2018

Microsoft Research, Redmond, WA

Research Intern, Human-Computer Interaction Group

Mentor: Steven Drucker

Designed, developed, and deployed interactive interface for operationalizing machine learning interpretability, published at CHI 2019.

Summer 2017

NASA Jet Propulsion Lab, Pasadena, CA

Creative Computer Scientist, Human Interfaces Group

Mentor: Scott Davidoff, Arun Viswanathan

Joint work between NASA JPL, Caltech, and Art Center creating interactive data visualizations for current scientific research. Prototype presented to lab leadership and secured funding to be incorporated into Mars 2020 mission.

Summer 2016

Pacific Northwest National Lab, Richland, WA

National Security Ph.D. Intern, Data Sciences & Analytics Group

Mentor: Nathan Hodas

Built interactive tools that generate synthetic images to explain deep learning classifiers, published at CHI 2017.

Academic Research Experience

Aug. 2016 — Dec. 2020

Georgia Institute of Technology, Atlanta, GA

Graduate Research Assistant, School of Computational Science and Engineering

Advisor: Duen Horng (Polo) Chau, Alex Endert

Member of the Polo Club of Data Science where we bridge and innovate at the intersection of data mining and human-computer interaction to synthesize scalable, interactive, and interpretable tools that amplify human's ability to understand and interact with big data.

Aug. 2015 — May 2016

Georgia Institute of Technology, Atlanta, GA

Graduate Research Assistant, School of Computational Science and Engineering

Mentor: Surya Kalidindi

Conducted research in physical data science and material informatics by creating property-structure linkages using machine learning to predict material properties. Contributed to direction and code of PyMKS: Materials Knowledge Systems in Python.

Jan. 2013 — May 2015

University of Georgia, Athens, GA

Undergraduate Research Assistant, Department of Mathematics

Advisor: David Gay

Explored 3D printing and mathematical exposition in topology. Programmed, designed, and 3D printed 34-piece, color-coordinated, and magnetized 3D puzzle of the trefoil knot fibration. Led 3D printing research and education in mathematics department.

Summer 2014

REU in Mathematics and Computational Science, Fairfield, CT

Fairfield University, Department of Mathematics

Mentor: Shanon Reckinger

Directly compared numerical solutions from Navier-Stokes equations to designed lab-scale experiments to model specific ocean phoneme. Configured MIT General Circulation Model on CPU cluster to run parallel computational fluid dynamics simulations.

Honors and Awards

- 2025 **Best Paper, Honorable Mention at ACM CHI Conference**
For "Exploring Empty Spaces: Human-in-the-Loop Data Augmentation"
- 2024 **Best Paper, Honorable Mention at ACM CHI Conference**
For "Talaria: Interactively Optimizing Machine Learning Models for Efficient Inference"
- 2023 **Best Paper, Honorable Mention at ACM IDC Conference**
For "Collaborative Machine Learning Model Building with Families Using Co-ML"
- 2022 **ACM SIGCHI Outstanding Dissertation Award**
For my dissertation "Interactive Scalable Interfaces for Machine Learning Interpretability"
- 2022 **Georgia Tech College of Computing Dissertation Award**
For my dissertation "Interactive Scalable Interfaces for Machine Learning Interpretability"
- 2022 **Best Paper at ACM CHI Conference**
For "Neo: Generalizing Confusion Matrix Visualization to Hierarchical and Multi-Output Labels"
- 2019 **Best Paper at ACM CHI Conference**
For "Managing Messes in Computational Notebooks"
- 2018 **Best Paper, Honorable Mention at VISxAI Workshop at IEEE VIS**
For "The Beginner's Guide to Dimensionality Reduction"
- 2018 — 2021 **NASA Space Technology Research Fellowship**
For my Ph.D. work on "Understanding Deep Neural Networks Through Attribution and Interactive Experimentation"
- 2018 **Audience Appreciation Award, Runner Up at ACM SIGKDD Conference**
For "Shield: Fast, Practical Defense and Vaccination for Deep Learning using JPEG Compression"
- 2017 — 2018 **Microsoft Azure for Research Award: AI for Earth**
For our work on "Deep Learning for Fine-scale Population Maps"
- 2017 **Best Demo, Honorable Mention at ACM SIGMOD/PODS Conference**
For "Visual Graph Query Construction and Refinement"
- 2015 — 2019 **President's Fellowship at Georgia Institute of Technology**

Select number of 1st year doctoral students who bring exemplary levels of scholarship and innovation to their academic departments

- 2015 **Outstanding Poster at JMM Undergraduate Poster Session in Computational Math**
For "Experimental and Numerical Comparison of Oceanic Overflow"
- 2015 **UGA CURO Research Graduation Distinction**
Awarded to undergraduates who write a thesis, present at the CURO Symposium, and complete 9 research credit hours
- 2014 **UGA CURO Research Assistantship**
Stipend awarded to outstanding undergraduates that actively participate in faculty-mentored research
- 2011 — 2015 **Dean's List**
Achieved at least a 3.5 GPA during a semester with minimum 14 credit hours
- 2011 — 2015 **Georgia HOPE Scholarship**
Merit-based award to Georgia residents providing tuition assistance for their undergraduate degree
- 2011 **Mission of Blessed Trinity: Artistic Sensibility**
One of two students to receive the Mission Statement award upon high-school graduation

Publications

Selected: Latest & Greatest

Semantic Regexes: Auto-Interpreting LLM Features with a Structured Language

Angie Boggust, Donghao Ren, Yannick Assogba, Dominik Moritz, Arvind Satyanarayan, Fred Hohman
International Conference on Learning Representations (ICLR). 2026.

[Project](#) [Demo](#) [PDF](#) [Code](#) [BibTeX](#) [DOI](#)

Model Compression in Practice: Lessons Learned from Practitioners Creating On-device Machine Learning Experiences

Fred Hohman, Mary Beth Kery, Donghao Ren, Dominik Moritz
ACM Conference on Human Factors in Computing Systems (CHI). Honolulu, Hawai'i, USA, 2024.

[Project](#) [PDF](#) [Video](#) [Preview](#) [BibTeX](#) [DOI](#)

Talaria: Interactively Optimizing Machine Learning Models for Efficient Inference

Fred Hohman, Chaoqun Wang, Jinmook Lee, Jochen Görtler, Dominik Moritz, Jeffrey P. Bigham, Zhile Ren, Cecile Foret, Qi Shan, Xiaoyi Zhang
ACM Conference on Human Factors in Computing Systems (CHI). Honolulu, Hawai'i, USA, 2024.

[Project](#) [PDF](#) [Video](#) [Preview](#) [BibTeX](#) [DOI](#) 🏆 **Best Paper, Honorable Mention**

Neo: Generalizing Confusion Matrix Visualization to Hierarchical and Multi-Output Labels

Jochen Görtler, Fred Hohman, Dominik Moritz, Kanit Wongsuphasawat, Donghao Ren, Rahul Nair, Marc Kirchner, Kayur Patel
ACM Conference on Human Factors in Computing Systems (CHI). New Orleans, LA, USA, 2022.

[Project](#) [Demo](#) [PDF](#) [Preview](#) [Recording](#) [Code](#) [BibTeX](#) [DOI](#) 🏆 **Best Paper**

Communicating with Interactive Articles

Fred Hohman, Matthew Conlen, Jeffrey Heer, Duen Horng (Polo) Chau
Distill (Distill). 2020.

[Project](#) [Demo](#) [PDF](#) [Code](#) [BibTeX](#) [DOI](#) ★ **Invited Commentary Interactive Article, Top of Hacker News**

Gamut: A Design Probe to Understand How Data Scientists Understand Machine Learning Models

Fred Hohman, Andrew Head, Rich Caruana, Robert DeLine, Steven Drucker
ACM Conference on Human Factors in Computing Systems (CHI). Glasgow, UK, 2019.

[Project](#) [Demo](#) [PDF](#) [Blog](#) [Video](#) [Preview](#) [Slides](#) [Code](#) [BibTeX](#) [DOI](#)

★ **Deployed at Microsoft Research**

Tech Report

TR2 **Apple Intelligence Foundation Language Models: Tech Report 2025**

Apple
arXiv:2507.13575 (arXiv). 2025.

[Project](#) [Demo](#) [PDF](#) [Blog](#) [BibTeX](#) [DOI](#)

TR1 **Apple Intelligence Foundation Language Models**

Apple
arXiv:2407.21075 (arXiv). 2024.

Journal

- J7 **Compress and Compare: Interactively Evaluating Efficiency and Behavior Across ML Model Compression Experiments**
Angie Boggust*, Venkatesh Sivaraman*, Yannick Assogba, Donghao Ren, Dominik Moritz, Fred Hohman
IEEE Transactions on Visualization and Computer Graphics (TVCG). St. Pete Beach, Florida, USA, 2025.
[Project](#) [PDF](#) [Video](#) [Code](#) [BibTeX](#) [DOI](#) *Authors contributed equally
- J6 **NeuroCartography: Scalable Automatic Visual Summarization of Concepts in Deep Neural Networks**
Haekyu Park, Nilaksh Das, Rahul Duggal, Austin P. Wright, Omar Shaikh, Fred Hohman, Duen Horng (Polo) Chau
IEEE Transactions on Visualization and Computer Graphics (TVCG). New Orleans, LA, USA, 2022.
[Project](#) [Demo](#) [PDF](#) [Recording](#) [Code](#) [BibTeX](#) [DOI](#) ★ **Top 4 TVCG Papers invited to SIGGRAPH**
- J5 **Communicating with Interactive Articles**
Fred Hohman, Matthew Conlen, Jeffrey Heer, Duen Horng (Polo) Chau
Distill (Distill). 2020.
[Project](#) [Demo](#) [PDF](#) [Code](#) [BibTeX](#) [DOI](#) ★ **Invited Commentary Interactive Article, Top of Hacker News**
- J4 **CNN Explainer: Learning Convolutional Neural Networks with Interactive Visualization**
Zijie J. Wang, Robert Turko, Omar Shaikh, Haekyu Park, Nilaksh Das, Fred Hohman, Minsuk Kahng, Duen Horng (Polo) Chau
IEEE Transactions on Visualization and Computer Graphics (TVCG). Salt Lake City, UT, USA, 2021.
[Project](#) [Demo](#) [PDF](#) [Video](#) [Recording](#) [Code](#) [BibTeX](#) [DOI](#)
★ **Top of GitHub Trending, Top 4 TVCG Papers invited to SIGGRAPH**
- J3 **Summit: Scaling Deep Learning Interpretability by Visualizing Activation and Attribution Summarizations**
Fred Hohman, Haekyu Park, Caleb Robinson, Duen Horng (Polo) Chau
IEEE Transactions on Visualization and Computer Graphics (TVCG). Vancouver, Canada, 2020.
[Project](#) [Demo](#) [PDF](#) [Video](#) [Recording](#) [Slides](#) [Code](#) [BibTeX](#) [DOI](#)
- J2 **Visual Analytics in Deep Learning: An Interrogative Survey for the Next Frontiers**
Fred Hohman, Minsuk Kahng, Robert Pienta, Duen Horng (Polo) Chau
IEEE Transactions on Visualization and Computer Graphics (TVCG). Berlin, Germany, 2018.
[Project](#) [Demo](#) [PDF](#) [Blog](#) [Video](#) [Slides](#) [Code](#) [BibTeX](#) [DOI](#)
- J1 **VIGOR: Interactive Visual Exploration of Graph Query Results**
Robert Pienta, Fred Hohman, Alex Endert, Acar Tamersoy, Kevin Roundy, Chris Gates, Shamkant Navathe, Duen Horng (Polo) Chau
IEEE Transactions on Visualization and Computer Graphics (TVCG). Phoenix, AZ, USA, 2018.
[Project](#) [PDF](#) [Video](#) [Preview](#) [BibTeX](#) [DOI](#)

Conference

- C23 **Understanding Annotator Decision-making with Interpretability**
Alex Oesterling, Donghao Ren, Yannick Assogba, Dominik Moritz, Sunnie S. Y. Kim, Leon Gatys, Fred Hohman
ACM Conference on Fairness, Accountability, and Transparency (FAccT). 2026.
→ More coming soon! [Project](#) [BibTeX](#) [DOI](#)
- C22 **Semantic Regexes: Auto-Interpreting LLM Features with a Structured Language**
Angie Boggust, Donghao Ren, Yannick Assogba, Dominik Moritz, Arvind Satyanarayan, Fred Hohman
International Conference on Learning Representations (ICLR). 2026.
[Project](#) [Demo](#) [PDF](#) [Code](#) [BibTeX](#) [DOI](#)
- C21 **Embedding Atlas: Low-Friction, Interactive Embedding Visualization**
Donghao Ren*, Fred Hohman*, Halden Lin, Dominik Moritz
IEEE Visualization Conference (VIS). 2025.
[Project](#) [Demo](#) [PDF](#) [Code](#) [BibTeX](#) [DOI](#) 🏆 **Top of GitHub Trending** *Authors contributed equally
- C20 **Policy Maps: Tools for Guiding the Unbounded Space of LLM Behaviors**
Michelle S. Lam, Fred Hohman, Dominik Moritz, Jeffrey P. Bigham, Kenneth Holstein, Mary Beth Kery
ACM Symposium on User Interface Software and Technology (UIST). 2025.

[Project](#) [PDF](#) [Code](#) [BibTeX](#) [DOI](#)

C19 **Exploring Empty Spaces: Human-in-the-Loop Data Augmentation**

Catherine Yeh, Donghao Ren, Yannick Assogba, Dominik Moritz, Fred Hohman

ACM Conference on Human Factors in Computing Systems (CHI). Yokohama, Japan, 2025.

[Project](#) [PDF](#) [Code](#) [BibTeX](#) [DOI](#) 🏆 **Best Paper, Honorable Mention**

C18 **Biscuit: Scaffolding LLM-Generated Code with Ephemeral UIs in Computational Notebooks**

Ruijia Cheng, Titus Barik, Alan Leung, Fred Hohman, Jeffrey Nichols

IEEE Symposium on Visual Languages and Human-Centric Computing (VL/HCC). 2024.

[Project](#) [PDF](#) [BibTeX](#) [DOI](#)

C17 **Model Compression in Practice: Lessons Learned from Practitioners Creating On-device Machine Learning Experiences**

Fred Hohman, Mary Beth Kery, Donghao Ren, Dominik Moritz

ACM Conference on Human Factors in Computing Systems (CHI). Honolulu, Hawai'i, USA, 2024.

[Project](#) [PDF](#) [Video](#) [Preview](#) [BibTeX](#) [DOI](#)

C16 **Talaria: Interactively Optimizing Machine Learning Models for Efficient Inference**

Fred Hohman, Chaoqun Wang, Jinmook Lee, Jochen Görtler, Dominik Moritz, Jeffrey P. Bigham, Zhile Ren, Cecile Foret, Qi Shan, Xiaoyi Zhang

ACM Conference on Human Factors in Computing Systems (CHI). Honolulu, Hawai'i, USA, 2024.

[Project](#) [PDF](#) [Video](#) [Preview](#) [BibTeX](#) [DOI](#) 🏆 **Best Paper, Honorable Mention**

C15 **Collaborative Machine Learning Model Building with Families Using Co-ML**

Tiffany Tseng, Jennifer King Chen, Mona Abdelrahman, Mary Beth Kery, Fred Hohman, Adriana Hilliard, R. Benjamin Shapiro

ACM Interaction Design and Children Conference (IDC). 2023.

[Project](#) [PDF](#) [Preview](#) [BibTeX](#) [DOI](#) 🏆 **Best Paper, Honorable Mention**

C14 **Angler: Helping Machine Translation Practitioners Prioritize Model Improvements**

Samantha Robertson*, Zijie J. Wang*, Dominik Moritz, Mary Beth Kery, Fred Hohman

ACM Conference on Human Factors in Computing Systems (CHI). Hamburg, Germany, 2023.

[Project](#) [Demo](#) [PDF](#) [Recording](#) [Code](#) [BibTeX](#) [DOI](#) *Authors contributed equally

C13 **Symphony: Composing Interactive Interfaces for Machine Learning**

Alex Bäuerle*, Ángel Alexander Cabrera*, Fred Hohman, Megan Maher, David Koski, Xavier Suau, Titus Barik, Dominik Moritz

ACM Conference on Human Factors in Computing Systems (CHI). New Orleans, LA, USA, 2022.

[Project](#) [Demo](#) [PDF](#) [Preview](#) [Recording](#) [Code](#) [BibTeX](#) [DOI](#) *Authors contributed equally

C12 **Neo: Generalizing Confusion Matrix Visualization to Hierarchical and Multi-Output Labels**

Jochen Görtler, Fred Hohman, Dominik Moritz, Kanit Wongsuphasawat, Donghao Ren, Rahul Nair, Marc Kirchner, Kayur Patel

ACM Conference on Human Factors in Computing Systems (CHI). New Orleans, LA, USA, 2022.

[Project](#) [Demo](#) [PDF](#) [Preview](#) [Recording](#) [Code](#) [BibTeX](#) [DOI](#) 🏆 **Best Paper**

C11 **Bluff: Interactively Deciphering Adversarial Attacks on Deep Neural Networks**

Nilaksh Das*, Haekyu Park*, Zijie J. Wang, Fred Hohman, Robert Firstman, Emily Rogers, Duen Horng (Polo) Chau

IEEE Visualization Conference (VIS). Salt Lake City, UT, USA, 2020.

[Project](#) [Demo](#) [PDF](#) [Recording](#) [Code](#) [BibTeX](#) [DOI](#) *Authors contributed equally

C10 **mage: Fluid Moves Between Code and Graphical Work in Computational Notebooks**

Mary Beth Kery, Donghao Ren, Fred Hohman, Dominik Moritz, Kanit Wongsuphasawat, Kayur Patel

ACM Symposium on User Interface Software and Technology (UIST). Minneapolis, MN, USA, 2020.

[Project](#) [PDF](#) [BibTeX](#) [DOI](#)

C9 **Understanding and Visualizing Data Iteration in Machine Learning**

Fred Hohman, Kanit Wongsuphasawat, Mary Beth Kery, Kayur Patel

ACM Conference on Human Factors in Computing Systems (CHI). Honolulu, HI, USA, 2020.

[Project](#) [PDF](#) [Video](#) [Preview](#) [Recording](#) [Slides](#) [BibTeX](#) [DOI](#)

C8 **FairVis: Visual Analytics for Discovering Intersectional Bias in Machine Learning**

Ángel Alexander Cabrera, Will Epperson, Fred Hohman, Minsuk Kahng, Jamie Morgenstern, Duen Horng (Polo) Chau

IEEE Conference on Visual Analytics Science and Technology (VAST). Vancouver, Canada, 2019.

[Project](#) [Demo](#) [PDF](#) [Blog](#) [Recording](#) [Slides](#) [Code](#) [BibTeX](#) [DOI](#)

C7 **TeleGam: Combining Visualization and Verbalization for Interpretable Machine Learning**

Fred Hohman*, Arjun Srinivasan*, Steven Drucker

IEEE Visualization Conference (VIS). Vancouver, Canada, 2019.

[Project](#) [Demo](#) [PDF](#) [Preview](#) [Recording](#) [Slides](#) [Code](#) [BibTeX](#) [DOI](#) *Authors contributed equally

C6 **ElectroLens: Understanding Atomistic Simulations through Spatially-resolved Visualization of High-dimensional Features**

Xiangyun Lei, Fred Hohman, Duen Horng (Polo) Chau, Andrew Medford

IEEE Visualization Conference (VIS). Vancouver, Canada, 2019.

[Project](#) [PDF](#) [Code](#) [BibTeX](#) [DOI](#)

C5 **Gamut: A Design Probe to Understand How Data Scientists Understand Machine Learning Models**

Fred Hohman, Andrew Head, Rich Caruana, Robert DeLine, Steven Drucker

ACM Conference on Human Factors in Computing Systems (CHI). Glasgow, UK, 2019.

[Project](#) [Demo](#) [PDF](#) [Blog](#) [Video](#) [Preview](#) [Slides](#) [Code](#) [BibTeX](#) [DOI](#)

★ **Deployed at Microsoft Research**

C4 **Managing Messes in Computational Notebooks**

Andrew Head, Fred Hohman, Titus Barik, Steven Drucker, Robert DeLine

ACM Conference on Human Factors in Computing Systems (CHI). Glasgow, UK, 2019.

[Project](#) [Demo](#) [PDF](#) [Video](#) [Preview](#) [Slides](#) [Code](#) [BibTeX](#) [DOI](#) 🏆 **Best Paper**

★ **Deployed in VS Code**

C3 **Atlas: Local Graph Exploration in a Global Context**

James Abello*, Fred Hohman*, Varun Bezzam, Duen Horng (Polo) Chau

ACM Conference on Intelligent User Interfaces (IUI). Los Angeles, CA, USA, 2019.

[Project](#) [PDF](#) [Video](#) [Talk](#) [Slides](#) [Code](#) [BibTeX](#) [DOI](#) *Authors contributed equally

C2 **Scalable K-Core Decomposition for Static Graphs Using a Dynamic Graph Data Structure**

Alok Tripathy, Fred Hohman, Duen Horng (Polo) Chau, Oded Green

IEEE International Conference on Big Data (Big Data). Seattle, WA, USA, 2018.

[Project](#) [PDF](#) [BibTeX](#) [DOI](#)

C1 **Shield: Fast, Practical Defense and Vaccination for Deep Learning using JPEG Compression**

Nilaksh Das, Madhuri Shanbhogue, Shang-Tse Chen, Fred Hohman, Siwei Li, Li Chen, Michael E. Kounavis, Duen Horng (Polo) Chau

ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD). London, UK, 2018.

[Project](#) [PDF](#) [Video](#) [Code](#) [BibTeX](#) [DOI](#) 🏆 **Audience Appreciation Award, Runner Up**

Preprint

P2 **A Scalable Approach to Clustering Embedding Projections**

Donghao Ren, Fred Hohman, Dominik Moritz

arXiv:2504.07285 (arXiv). 2025.

[Project](#) [Demo](#) [PDF](#) [Code](#) [BibTeX](#) [DOI](#)

P1 **Keeping the Bad Guys Out: Protecting and Vaccinating Deep Learning with JPEG Compression**

Nilaksh Das, Madhuri Shanbhogue, Shang-Tse Chen, Fred Hohman, Li Chen, Michael E. Kounavis, Duen Horng (Polo) Chau

arXiv:1705.02900 (arXiv). 2017.

[Project](#) [PDF](#) [Code](#) [BibTeX](#)

Workshop

W7 **Designing Data: Proactive Data Collection and Iteration for Machine Learning Using Reflexive Planning, Monitoring, and Density Estimation**

Aspen Hopkins, Fred Hohman, Luca Zappella, Xavier Suau, Dominik Moritz

Artificial Intelligence & Human-Computer Interaction Workshop at ICML (AI&HCI). 2023.

[Project](#) [PDF](#) [Poster](#) [BibTeX](#)

W6 **Launching the Parametric Press**

Matthew Conlen, Fred Hohman

Visualization for Communication at IEEE VIS (VisComm). Vancouver, Canada, 2019.

[Project](#) [Demo](#) [PDF](#) [Code](#) [BibTeX](#)

W5 **Discovery of Intersectional Bias in Machine Learning Using Automatic Subgroup Generation**

Ángel Alexander Cabrera, Minsuk Kahng, Fred Hohman, Jamie Morgenstern, Duen Horng (Polo) Chau

Debugging Machine Learning Models Workshop at ICLR (Debug ML). New Orleans, LA, USA, 2019.

[Project](#) [PDF](#) [BibTeX](#)

W4 **The Beginner's Guide to Dimensionality Reduction**

Matthew Conlen, Fred Hohman

Workshop on Visualization for AI Explainability at IEEE VIS (VISxAI). Berlin, Germany, 2018.

[Project](#) [Demo](#) [Slides](#) [Code](#) [BibTeX](#) 🏆 **Best Paper, Honorable Mention**

W3 **Compression to the Rescue: Defending from Adversarial Attacks Across Modalities**

Nilaksh Das, Madhuri Shanbhogue, Shang-Tse Chen, Fred Hohman, Siwei Li, Li Chen, Michael E. Kounavis, Duen Horng (Polo) Chau

Project Showcase, ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD). London, UK, 2018.

[Project](#) [PDF](#) [Code](#) [BibTeX](#)

W2 **A Viz of Ice and Fire: Exploring Entertainment Video Using Color and Dialogue**

Fred Hohman, Sandeep Soni, Ian Stewart, John Stasko

2nd Workshop on Visualization for the Digital Humanities at IEEE VIS (VIS4DH). Phoenix, AZ, USA, 2017.

[Project](#) [Demo](#) [PDF](#) [Slides](#) [Code](#) [Data](#) [BibTeX](#)

W1 **A Deep Learning Approach for Population Estimation from Satellite Imagery**

Caleb Robinson, Fred Hohman, Bistra Dilkina

1st ACM SIGSPATIAL Workshop on Geospatial Humanities (GeoHum). Redondo Beach, CA, USA, 2017.

[Project](#) [Demo](#) [PDF](#) [Code](#) [BibTeX](#) 🏆 **Microsoft AI for Earth Award**

Poster

P6 **The Future of Notebook Programming Is Fluid**

Mary Beth Kery, Donghao Ren, Kanit Wongsuphasawat, Fred Hohman, Kayur Patel

Extended Abstracts on ACM Human Factors in Computing Systems (CHI). Honolulu, HI, USA, 2020.

[Project](#) [PDF](#) [BibTeX](#) [DOI](#)

P5 **CNN 101: Interactive Visual Learning for Convolutional Neural Networks**

Zijie J. Wang, Robert Turko, Omar Shaikh, Haekyu Park, Nilaksh Das, Fred Hohman, Minsuk Kahng, Duen Horng (Polo) Chau

Extended Abstracts on ACM Human Factors in Computing Systems (CHI). Honolulu, HI, USA, 2020.

[Project](#) [PDF](#) [Video](#) [Code](#) [BibTeX](#) [DOI](#)

P4 **Massif: Interactive Interpretation of Adversarial Attacks on Deep Learning**

Nilaksh Das*, Haekyu Park*, Zijie J. Wang, Fred Hohman, Robert Firstman, Emily Rogers, Duen Horng (Polo) Chau

Extended Abstracts on ACM Human Factors in Computing Systems (CHI). Honolulu, HI, USA, 2020.

[Project](#) [PDF](#) [BibTeX](#) [DOI](#) *Authors contributed equally

P3 **NeuralDivergence: Exploring and Understanding Neural Networks by Comparing Activation Distributions**

Haekyu Park, Fred Hohman, Duen Horng (Polo) Chau

Poster, IEEE Pacific Visualization Symposium (PacificVis). Bangkok, Thailand, 2019.

[Project](#) [Demo](#) [PDF](#) [Slides](#) [Poster](#) [BibTeX](#)

P2 **3D Exploration of Graph Layers via Vertex Cloning**

James Abello*, Fred Hohman*, Duen Horng (Polo) Chau

Poster, IEEE Conference on Visual Analytics Science and Technology (VAST). Phoenix, AZ, USA, 2017.

[Project](#) [PDF](#) [Video](#) [Poster](#) [BibTeX](#) *Authors contributed equally

P1 **ShapeShop: Towards Understanding Deep Learning Representations via Interactive Experimentation**

Fred Hohman, Nathan Hodas, Duen Horng (Polo) Chau

Extended Abstracts on ACM Human Factors in Computing Systems (CHI). Denver, CO, USA, 2017.

[Project](#) [PDF](#) [Video](#) [Poster](#) [Code](#) [BibTeX](#) [DOI](#)

Demo

- D4

WizMap: Scalable Interactive Visualization for Exploring Large Machine Learning Embeddings

Zijie J. Wang, Fred Hohman, Duen Horng (Polo) Chau

Demo, Annual Meeting of the Association for Computational Linguistics (ACL). Toronto, Canada, 2023.

[Project](#) [Demo](#) [PDF](#) [Video](#) [Code](#) [BibTeX](#) [DOI](#)
- D3

Interactive Classification for Deep Learning Interpretation

Ángel Alexander Cabrera, Fred Hohman, Jason Lin, Duen Horng (Polo) Chau

Demo, Conference on Computer Vision and Pattern Recognition (CVPR). Salt Lake City, UT, USA, 2018.

[Project](#) [Demo](#) [PDF](#) [Video](#) [Code](#) [BibTeX](#)
- D2

mHealth Visual Discovery Dashboard

Dezhi Fang, Fred Hohman, Peter Polack, Hillol Sarker, Minsuk Kahng, Moushumi Sharmin, Mustafa al'Absi, Duen Horng (Polo) Chau

Demo, ACM International Joint Conference on Pervasive and Ubiquitous Computing (Ubicomp). Maui, HI, USA, 2017.

[Project](#) [PDF](#) [Video](#) [Poster](#) [BibTeX](#) [DOI](#)
- D1

Visual Graph Query Construction and Refinement

Robert Pienta, Fred Hohman, Acar Tamersoy, Alex Endert, Shamkant Navathe, Hanghang Tong, Duen Horng (Polo) Chau

Demo, ACM International Conference on Management of Data (SIGMOD/PODS). Chicago, IL, USA, 2017.

[Project](#) [PDF](#) [Video](#) [Poster](#) [BibTeX](#) [DOI](#) [🏆 Best Demo, Honorable Mention](#)

Miscellaneous

- M2

The Effect of Numerical Parameters on Eddies in Oceanic Overflows: A Laboratory and Numerical Study

Shanon Reckinger, Thomas Gibson, Fred Hohman, Theresa Morrison, Scott Reckinger, Mateus Carvalho

International Journal of Computational Methods and Experimental Measurements (CMEM). 2019.

[Project](#) [PDF](#) [BibTeX](#)
- M1

Experimental and Numerical Comparison of Oceanic Overflow

Thomas Gibson, Fred Hohman, Theresa Morrison, Shanon Reckinger, Scott Reckinger

Abstract, American Physical Society Division of Fluid Dynamics (APS DFD). San Francisco, CA, USA, 2014.

[Project](#) [Blog](#) [Poster](#)

Talks

- Aug. 2024

Tiny but Powerful: Human-Centered Research to Support Efficient On-Device ML

Apple Workshop on Human-Centered Machine Learning
- Nov. 2022

Interactive Scalable Interfaces for Machine Learning Interpretability

Northeastern University, Visualization for Machine Learning Course Guest Lecture
- May 2022

ACM Conference on Human Factors in Computing Systems
- Apr. 2022

Carnegie Mellon University, Interactive Data Science Course Guest Lecture
- Nov. 2021

Carnegie Mellon University, Data Visualization Course Guest Lecture
- Oct. 2020

University of Wisconsin-Madison, Data Visualization Course Guest Lecture
- Oct. 2020

Georgia Tech Thesis Defense
- May 2020

IBM Research
- Apr. 2020

Microsoft Research
- Apr. 2020

Apple
- Apr. 2020

Autodesk Research
- Dec. 2019

Georgia Tech Thesis Proposal
- May 2022

Neo: Generalizing Confusion Matrix Visualization to Hierarchical and Multi-Output Labels

ACM Conference on Human Factors in Computing Systems
- Mar. 2020

Summit: Scaling Deep Learning Interpretability by Visualizing Activation and Attribution Summarizations

NVIDIA GTC
- Oct. 2019

IEEE Visualization
- Oct. 2019

TeleGam: Combining Visualization and Verbalization for Interpretable Machine Learning

IEEE Visualization

	Gamut: A Design Probe to Understand How Data Scientists Understand Machine Learning Models
June 2019	Microsoft Machine Learning and Data Science Summit
May 2019	ACM Conference on Human Factors in Computing Systems
	Explaining Machine Learning Models Using Interactive Visualization
Mar. 2019	Georgia Tech School of CSE Strategic Partnership Program Summit
Apr. 2019	Georgia Tech CSE 6242 Data and Visual Analytics
Mar. 2019	Symantec Research Labs
Mar. 2019	NASA Jet Propulsion Laboratory
	Atlas: Local Graph Exploration in a Global Context
Mar. 2019	ACM Intelligent User Interfaces
	Visual Analytics in Deep Learning: An Interrogative Survey for the Next Frontiers
Jan. 2019	Carnegie Mellon University
Oct. 2018	University of Georgia
Oct. 2018	IEEE Visualization
	The Beginner's Guide to Dimensionality Reduction
Oct. 2018	VISxAI Workshop at IEEE Visualization
	Comparing Interactive Local and Global Explanation Paradigms for Human-assisted Machine Learning Tasks
July 2018	Microsoft Research
	Graph Playgrounds: 3D Exploration of Graph Layers via Vertex Cloning
Dec. 2017	AT&T Research Labs Graduate Student Symposium
	A Viz of Ice and Fire: Exploring Entertainment Video Using Color and Dialogue
Oct. 2017	2nd Workshop on Visualization for the Digital Humanities at IEEE Visualization
	Constellation: Visualizing Cybersecurity in Real Time
Aug. 2017	NASA Jet Propulsion Laboratory
Aug. 2017	California Institute of Technology
	Visualizing Learned Semantics with Deep Learning
Nov. 2016	Georgia Tech Ph.D. Qualifying Oral Exam
	Drawing Semantics with Deep Learning
2016	Pacific Northwest National Laboratory
	3D Printing The Trefoil Knot And Its Pages
Mar. 2015	UGA Center for Undergraduate Research Symposium, included hands-on demo
	Experimental and Numerical Studies of Oceanic Overflow
June 2015	AMS Conference on Atmospheric and Oceanic Fluid Dynamics
Jan. 2015	Joint Mathematics Meeting
Nov. 2014	APS Division of Fluid Dynamics
Aug. 2014	Invited and presented on behalf at Brown University, Los Alamos National Lab
July 2014	Northeast REU Mini-Conference at Yale University
July 2014	University of Rhode Island Bay Campus
	3D Printing in Topology
Mar. 2014	UGA Center for Undergraduate Research Symposium, included hands-on demo

Press

June 2022	"Alumnus Building Legacy through Dissertation and Mentorship," Georgia Tech, College of Computing
Nov. 2020	"How interactive, visual storytelling helps readers better retain information," Storybench
Nov. 2020	"Dense Discovery Issue 112," Dense Discovery
Oct. 2020	"Numlock Sunday: Halden Lin and Aatish Bhatia on the Parametric Press Climate Issue," Numlock News
Oct. 2020	"Web Archives as Digital Publications / Digital Publications as Web Archives," Webrecorder
Oct. 2020	"Interactive data essays on climate change," Flowing Data
July 2020	"Python in Visual Studio Code - VSCode Features You Need to Know," Tech With Tim

July 2020	"Python in Visual Studio Code – July 2020 Release," Microsoft Developer Blog
June 2020	"How Do Neural Networks Learn?," Two Minute Papers
Mar. 2020	"Visualizing Fairness in Machine Learning," Data Stories Podcast
Nov. 2019	"The Interactive News Platform for Everyone," Stack Overflow Blog
Oct. 2019	"Is this the dynamic web we were promised?," Hanselminutes Podcast
June 2019	"Myth of the impartial machine," Flowing Data
May 2019	"The Secret Life of a JPEG," Fast Company
Dec. 2018	"'Human Rights' May Help Shape Artificial Intelligence in 2019," Georgia Tech, College of Computing
Dec. 2018	"Designers, Programmers, and Researchers Join Forces to Create a New Kind of Digital Magazine Called the Parametric Press," Georgia Tech, College of Computing
June 2018	"Georgia Tech Teams up with Intel to Protect Artificial Intelligence from Malicious Attacks Using SHIELD," Georgia Tech, College of Computing
Dec. 2017	"Georgia Tech Team To Use Microsoft Grant to Study Human Migration Dynamics," Georgia Tech, College of Computing
Sept. 2015	"Georgia Tech PhD Student Puts Finishing Touches on 3D Printed Trumpety Trefoil," 3dprint.com
Spring 2015	"Student Profile: Fred Hohman," 2015 UGA Mathematics Department Newsletter
Feb. 2015	"Falling Water," MITgcm.org
Dec. 2014	"Mathematics/Physics Student Creates 3D Printed Puzzle of Trefoil Knot, Catches Mathematical Community's Interest," 3dprint.com
July 2014	"Day 311 - Trefoil Trumpet," Makerhome.com
Apr. 2014	"Mathematics with 3D Printing," Sketches of Topology

Teaching

Spring 2019	Graduate Teaching Assistant <i>Georgia Institute of Technology, Atlanta, GA</i> Information Visualization (CS 4460), Instructor: Alex Endert Designed homeworks, held weekly office hours, and mentored student team projects for Information Visualization (CS 4460), an undergraduate course with 134 students enrolled.
Spring 2017	Graduate Teaching Assistant <i>Georgia Institute of Technology, Atlanta, GA</i> Data and Visual Analytics (CSE 6242 / CX 4242), Instructor: Duen Horng (Polo) Chau Designed homeworks, held weekly office hours, and mentored student team projects for Data and Visual Analytics (CSE 6242 / CX 4242), a graduate course with 214 students enrolled.
2014 — 2015	Student Notetaker <i>University of Georgia, Athens, GA</i> Generated notes for undergraduate mathematics and physics courses for students with disabilities.
2012	Tutor <i>University of Georgia, Athens, GA</i> Specialized in tutoring calculus to undergraduates.

Mentoring

Summer 2025	Angie Boggust at Apple <i>Ph.D. in Computer Science, Massachusetts Institute of Technology</i> Auto interpretability for large language models 🧑 Apple Scholar, PhD Fellowship in Human Centered AI
Summer 2025	Alex Oesterling at Apple <i>Ph.D. in Computer Science, Harvard University</i> Understanding annotation differences with interpretability
Summer 2024	Catherine Yeh at Apple <i>Ph.D. in Computer Science, Harvard University</i> Interactive techniques for data augmentation
Summer 2024	Michelle S. Lam at Apple

	<i>Ph.D. in Computer Science, Stanford University</i> Tools for grounding LLM policy design
Summer 2023	Angie Boggust at Apple <i>Ph.D. in Computer Science, Massachusetts Institute of Technology</i> Machine learning model compression visualization 👤 Apple Scholar, PhD Fellowship in Human Centered AI
Summer 2023	Venkatesh Sivaraman at Apple <i>Ph.D. in Human-computer Interaction, Carnegie Mellon University</i> Machine learning model compression visualization
Summer 2023	Ruanqianqian Huang at Apple <i>Ph.D. in Computer Science, UC San Diego</i> Non-expert tools for machine learning
Summer 2022	Samantha Robertson at Apple <i>Ph.D. in Electrical Engineering & Computer Sciences, UC Berkeley</i> Interactive tools for machine translation
Summer 2022	Zijie Jay Wang at Apple <i>Ph.D. in Machine Learning, Georgia Tech</i> Interactive tools for machine translation 👤 Apple Scholar, PhD Fellowship in Human Centered AI
Spring + Summer 2021	Alex Bäuerle at Apple <i>Ph.D. in Information Technology, Ulm University</i> Reusable, explorable, and sharable data science components
Summer 2021	Ángel Alexander Cabrera at Apple <i>Ph.D. in Human-computer Interaction, Carnegie Mellon University</i> Reusable, explorable, and sharable data science components
Summer 2020	Jochen Görtler at Apple <i>Ph.D. in Computer Science, University of Konstanz</i> Interactive and hierarchical confusion matrix visualization
Summer 2020	Aspen Hopkins at Apple <i>Ph.D. in Computer Science, Massachusetts Institute of Technology</i> Designing data and proactive data collection for machine learning
Fall 2019 — Fall 2020	Omar Shaikh <i>B.S. in Computer Science, Georgia Institute of Technology</i> Visualization for machine learning education 👤 Sigma Xi Best Undergraduate Research Award, Georgia Institute of Technology
Fall 2019 — Fall 2020	Robert Turko <i>B.S. in Computer Science, Georgia Institute of Technology</i> Visualization for machine learning education 👤 Outstanding Senior Award, College of Computing, Georgia Institute of Technology
Fall 2019 — Spr. 2020	Rob Firstman <i>B.S. in Computer Science, Georgia Institute of Technology</i> Visualization for deep learning interpretability
Spr. 2019 — Spr. 2020	Will Epperson <i>B.S. in Computer Science, Georgia Institute of Technology</i> Visual analytics for machine learning fairness 👤 Stamps President's Scholar Now: PhD Student (HCI) at Carnegie Mellon University
Spr. 2019 — Spr. 2020	Siwei Li <i>B.S. in Computer Science, Georgia Institute of Technology</i> Visual graph analytics 👤 Outstanding Undergraduate Researcher, College of Computing, Georgia Institute of Technology 👤 President's Undergraduate Research Award Now: Software Engineer II at Google
Spr. 2018 — Spr. 2019	Ángel Alexander Cabrera

B.S. in Computer Science, Georgia Institute of Technology

Visual analytics for machine learning fairness, interactive classification for deep learning

👤 National Science Foundation Graduate Research Fellowship Program (NSF GRFP)

👤 Love Family Foundation Scholarship (most outstanding graduating senior), Georgia Institute of Technology

👤 Stamps President's Scholar

Now: PhD Student (HCI) at Carnegie Mellon University

Fall 2016 — Spr. 2018

Dezhi Fang

B.S. in Computer Science, Georgia Institute of Technology

Interactive visual motif discovery

👤 Outstanding Undergraduate Researcher, College of Computing, Georgia Institute of Technology

👤 Faculty Materials, Supplies, and Travel Grants for Undergraduate Research

👤 Awarded President's Undergraduate Research Travel Award

Now: Software Development Engineer at Airbnb

Fall 2017 — Spr. 2018

Prasenjeet Biswal

M.S. in Computer Science, Georgia Institute of Technology

Visualization for deep learning attribution

Now: Software Development Engineer at Oath

Grants and Funding

2018 — 2021

Understanding Deep Neural Networks Through Attribution and Interactive Experimentation

NSTRF: NASA Space Technology Research Fellowship

Co-PIs: Duen Horng (Polo) Chau

Funded \$80,000/year for 3 years

2017 — 2018

Deep Learning for Fine-scale Population Maps

Microsoft Azure for Research Award: AI for Earth

Co-PIs: Caleb Robinson, Bistra Dilkina

Funded \$15,000

Fall 2014

3D Printing the Trefoil Knot and its Pages

UGA CURO Research Assistantship

Co-PIs: David Gay

Funded \$1,000

Interactive Articles

Sept. 2023

Menstrual Cycle Length and Variability: A Visual Explanation

Samantha Robertson, Halden Lin, Fred Hohman

Visualizing the Apple Women's Health Study

June 2022

Awesome Mathematical Notation Design

Fred Hohman

List of links of techniques to help augment mathematical notation

May 2021 — Present

Apple Machine Learning Research Articles

Interactively explore advancements in machine learning research at Apple

2026

ParaRNN: Large-Scale Nonlinear RNNs, Trainable in Parallel

2025

Exploring LLMs with MLX and the Neural Accelerators in the M5 GPU

The "Super Weight:" How Even a Single Parameter can Determine a Large Language Model's Behavior

FastVLM: Efficient Vision Encoding for Vision Language Models

Updates to Apple's On-Device and Server Foundation Language Models

An LLM-Based Approach to Review Summarization on the App Store

Understanding Aggregate Trends for Apple Intelligence Using Differential Privacy

Controlling Language and Diffusion Models by Transporting Activations

2024

Accelerating LLM Inference on NVIDIA GPUs with ReDrafter

On Device Llama 3.1 with Core ML

Combining Machine Learning and Homomorphic Encryption in the Apple Ecosystem

Improving How Machine Translations Handle Grammatical Gender Ambiguity

Introducing Apple's On-Device and Server Foundation Models

Enhancing Paragraph Generation with a Latent Language Diffusion Model

	Personalizing Health and Fitness with Hybrid Modeling Humanizing Word Error Rate for ASR Transcript Readability and Accessibility Deploying Attention-Based Vision Transformers to Apple Neural Engine
2023	Advancing Speech Accessibility with Personal Voice Voice Trigger System for Siri Learning Iconic Scenes with Differential Privacy Fast Class-Agnostic Salient Object Segmentation Improved Speech Recognition for People Who Stutter
2022	3D Parametric Room Representation with RoomPlan A Multi-Task Neural Architecture for On-Device Scene Analysis Deploying Transformers on the Apple Neural Engine Modeling the Impact of User Mobility on COVID-19 Infection Rates Over Time
2021	Interpretable Adaptive Optimization On-device Panoptic Segmentation for Camera Using Transformers Improving Neural Network Subspaces Learning to Generate Radiance Fields of Indoor Scenes Recognizing People in Photos Through Private On-Device Machine Learning Making Mobile Applications Accessible with Machine Learning
Oct 2020	Parametric Press, Issue 02: The Climate Issue Matthew Conlen, Fred Hohman, Sara Stalla, Andrew Sass Interactively analyze our climate's past, present, and future Your Personal Carbon History by Aatish Bhatia Why you've lived through a period of unprecedented carbon emissions. The Corporations Behind Climate Change by Geoffrey Litt, Seth Thompson Seven simple demands to hold fossil fuel companies accountable. Tiny Algae and the Political Theater of Planting One Trillion Trees by Benjamin Cooley To fight climate change, it's time to start thinking big by thinking small. Drought of the Sinking Delta by Christina Orieschnig Things are out of joint in the delta of the Mekong, Southeast Asia's largest river. The Hidden Cost of Digital Consumption by Halden Lin, Aishwarya Nirmal, Shobhit Hathi, Lilian Liang Digital doesn't mean green. How much carbon dioxide was released when you loaded this article?
May 2019	Parametric Press, Issue 01: Science + Society Matthew Conlen, Fred Hohman, Sara Stalla, Victoria Uren, Andrew Sass Examine scientific and technological phenomena that stand to shape society at large, ★ Top of Hacker News, twice Unraveling the JPEG by Omar Shehata JPEG images are everywhere, but behind them are algorithms that remove details that are imperceptible to the human eye. The Myth of the Impartial Machine by Alice Feng, Shuyan Wu Machine learning models reproduce inequalities within data they're fed. Can programmers free their models from prejudice? Data Science for Fair Housing by Alyson Powell Key In Atlanta, how effective is a new nonprofit's plan to resist displacement by supporting the city's most vulnerable residents? Flatland Follies: An Adjunct Simulator by James McGirk Play your way through the daily struggles of teaching and preserving art at a declining college. On Particle Physics by Riccardo Maria Bianchi A CERN particle physicist walks through the history and science of particle physics, and why you should care about it. Anything That Flies, On Anything That Moves by Megan Vo Dig into data of over two million bombings during a covert US assault on Southeast Asian countries from the 1960s and 70s.
Nov. 2018	Blueberry Pancakes Caleb Robinson, Fred Hohman A toy algorithms problem
July 2018	The Beginner's Guide to Dimensionality Reduction Matthew Conlen, Fred Hohman Explore the methods data scientists use to visualize high-dimensional data, ★ VISxAI Best Paper, Honorable Mention
June 2018	The Math of Card Shuffling Fred Hohman Riffing from factory order to complete randomness, ★ Top of Hacker News, twice
Oct. 2017	A Viz of Ice and Fire

Service

Organizer

Workshop on Visualization for AI Explainability (**VISxAI**) at IEEE VIS 2024, 2023, 2022, 2021, 2020, 2019

Program Committee

IEEE Visualization (**VIS**) 2025, 2024, 2023

ACM International Conference on Intelligent User Interfaces (**IUI**) 2023, 2022, 2021, 2020, 2019

Debugging Machine Learning Models Workshop (**DebugML**) at ICLR 2019

Symposium on Visualization in Data Science (**VDS**) at IEEE VIS 2022, 2018

Workshop on Visualization for AI Explainability (**VISxAI**) at IEEE VIS 2018

Workshop on Interactive Data Exploration and Analytics (**IDEA**) at KDD 2018

Reviewer

ACM Conference on Human Factors in Computing Systems (**CHI**) 2024, 2023, 2022, 2021, 2020, 2019, 2018, 2017

IEEE Visualization (**VIS**) 2025, 2024, 2023, 2022, 2021, 2020, 2019, 2018, 2017

ACM Transactions on Computer-Human Interaction (**TOCHI**) 2023

PacificVis Visual Data Storytelling (**VisStory**) 2023

ACM Conference on Fairness, Accountability, and Transparency (**FAccT**) 2022

Visualization Meets AI at PacificVis 2022

Rethinking ML Papers at ICLR 2021

ACM Designing Interactive Systems (**DIS**) 2021

ACM Transactions on Interactive Intelligent Systems (**TIIS**) 2021

IEEE Pacific Visualization Conference (**PacificVis**) 2021

ACM User Interface Software and Technology Symposium (**UIST**) 2021, 2020

Symposium on Visualization in Data Science (**VDS**) at IEEE VIS 2022, 2020, 2019, 2018

Distill Research Journal (**Distill**) 2019

ACM SIGKDD Conference on Knowledge Discovery and Data Mining (**KDD**) 2019, 2017

ACM Conference on Computer Supported Cooperative Work and Social Computing (**CSCW**) 2019

Human-Centered Machine Learning Perspectives Workshop (**HCMLP**) at CHI 2019

1st Deep Learning and Security Workshop (**DLS**) at IEEE SP 2018

IEEE International Conference on Distributed Computing Systems (**ICDCS**) 2017

SIAM International Conference on Data Mining (**SDM**) 2017

Institutional

Georgia Tech CSE Graduate Student Association, Vice President, 2018 – 2020

Georgia Tech CSE Chair Search Committee, 2019 – 2020

Member

2016 — Present	Association for Computing Machinery (ACM)
2016 — Present	Institute of Electrical and Electronics Engineers (IEEE)
2012 — 2015	UGA Mathematics Club
2012 — 2013	Society of Physics Students, UGA Chapter (SPS)
2011 — 2015	National Society of Collegiate Scholars (NSCS)

Design

2025 Generative AI

Apple Human Interface Guidelines

Generative AI empowers you to enhance your app or game with dynamic content and offer intelligent features that unlock new levels of creativity, connection, and productivity

2022 Charting Data

Apple Human Interface Guidelines

Presenting data in a chart can help you communicate information with clarity and appeal

2022 Charts

Apple Human Interface Guidelines

Organize data in a chart to communicate information with clarity and visual appeal

2017 — 2018

IDEA Workshop Proceedings Cover (2017, 2018)

ACM SIGKDD Workshop on Interactive Data Exploration and Analytics (IDEA)

Designed workshop poster and conference proceedings cover

2017, 2022, 2023

Brad Myers Advisee Tree

ACM Conference on Human Factors in Computing Systems (CHI), Denver, USA

Designed and implemented an interactive visualization of Brad Myers's advisee tree shown during his CHI 2017 Lifetime Research Award talk; designed accompanying ribbon worn by attendees at the conference

Aug. 2014

3D Printed Cube Decomposition Trophy

University of Georgia Mathematics Department, Athens, USA

Designed, modeled, and 3D printed cube decomposition trophy for annual UGA High School Math Tournament that was given to the top scoring teams and participants

Aug. 2014

3D Printed UGA Keychain

University of Georgia Lamar Dodd School of Art, Athens, USA

Created 3D printed UGA keychain and presentation notes given at Experience UGA: a interdisciplinary event that exposes middle-school and high-school students to hands-on learning activities

References

Dr. Polo Chau, Associate Professor

School of Computational Science and Engineering

Georgia Institute of Technology

cc.gatech.edu/~dchau/

Dr. Alex Endert, Associate Professor

School of Interactive Computing

Georgia Institute of Technology

va.gatech.edu/endert

Dr. Scott Davidoff, Senior Manager

Human-Centered Design Group

NASA Jet Propulsion Lab

scottdavidoff.com

Dr. Steven Drucker, Partner and Research Manager

Visualization and Interactive Data Analysis Group

Microsoft Research

microsoft.com/en-us/research/people/sdrucker

Dr. Nathan Hodas, Senior Research Scientist

Data Sciences and Analytics Group

Pacific Northwest National Laboratory

[linkedin.com/in/nathan-hodas-ba37956](https://www.linkedin.com/in/nathan-hodas-ba37956)